

EXPERIMENT NO: 11

STUDY OF GPIO & BIT ADDRESSING IN AT89C51

AIM:

To study GPIO control and bit-addressable SFR using Embedded C.

SOFTWARE REQUIRED:

- Keil

THEORY:

8051 ports are byte and bit addressable. Using sbit, individual bits of a port can be accessed directly. Ports default to logic HIGH due to internal pull-ups.

PROGRAM:

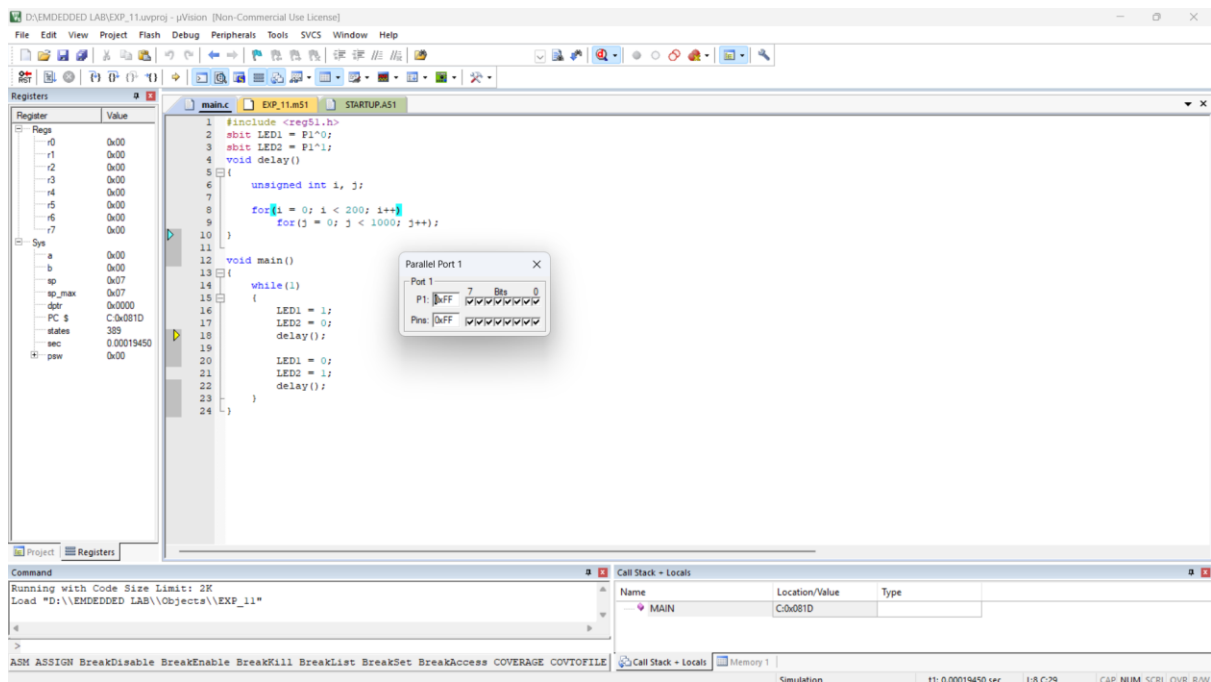
```
#include <reg51.h>

sbit LED1 = P1^0;
sbit LED2 = P1^1;

void delay(){
    unsigned int i,j;
    for(i=0;i<200;i++)
        for(j=0;j<1000;j++);}

void main(){
    while(1)    {
        LED1 = 1;
        LED2 = 0;
        delay();
        LED1 = 0;
        LED2 = 1;
        delay();    }
}
```

OUTPUT:



RESULT:

Thus GPIO and bit addressing was studied and in Port1 the pins **P1.0** and **P1.1** toggled alternately.